

AE-310: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h

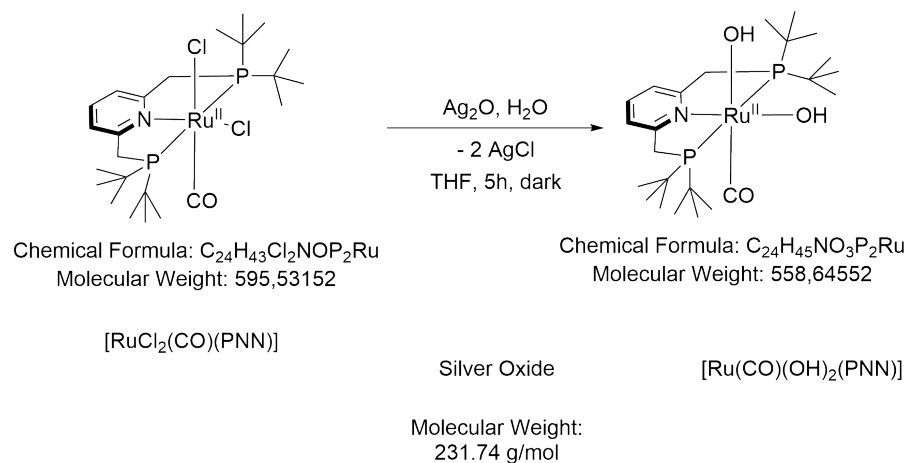
Date: 2024-07-02

Tags: Dihydroxo Ag₂O PNP
P(tBu)N(py)P(tBu) $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$
NMR AE 1H 31P

Status: Done

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Reaction scheme/sample structure



Literature/reference experiments

Literature	https://doi.org/10.1039/D1EE01053K
Reproduction	/
Related experiments	Experiment - AE-KRA-204: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h Experiment - AE-238: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h Experiment - AE-252: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h

Reagents

Name	CAS-Number/Expermient-Number	Amount [mmol]	Equivalen ts	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Volume _{theo} [ml]	Volume _{exp} [ml]	Density [g/mL]
$[\text{RuCl}_2(\text{CO})(\text{PNP}\{\text{tBu}\})]$	Experiment - AE-261: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{PNP})]$ with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$ (AE-261-1)	0.500	1.00	297.77	298.20	595.53	/	/	/
Ag_2O	20667-12-3	3.07	6.14	711.44	709.21	231.74	/	/	/
H_2O	Experiment - KRA-104: Degassing of Milli-Q water	123	234	2216.46	/	18.02	2.22	2.2	0.998
THF	Experiment - KRA-080: Degassing of THF	/	/	/	/	/	6.9	15 + 20	/
n-Heptane	Experiment - AE-121: Degassing of n-Heptane	/	/	/	/	/	/	20	/

Procedure/observations

All steps, unless mentioned otherwise, were carried out under argon using standard Schlenk technique or in an argon

filled glovebox. (see Protocol [Protocol - Schlenk Technique](#))

Due to the instability of Ag₂O and Dihydroxo to the light, the procedure was done mostly in the dark. The precise measures, which were taken, are listed in "**Steps**".

Date	Time	Step	Observations
02.07		A 25mL-brown glas Schlenk flask was prepared, by heating under vacuum.	
	9:00	[RuCl ₂ (CO)(PNP)] (298.20 mg) and Ag ₂ O (709.21 mg) was weighted in air into the reaction schlenk flask and the flask was set under argon.	
		The light in the fumehood was turned off	
	9:30 - 9:35	THF (15 mL) and water (2.2 mL) were added to the reaction schlenk flask.	
	9:35 - 14:35	The reaction mixture was stirred for 5 h at 1000 rpm.	
	14:35 - 14:37	The reaction mixture was allowed to settle	
	14:37 - 15:00	The mixture was filtered through a pore 4 frit according to Protocol - Filtration with frit technique into a 50 mL-Schlenk	filtrate: AE-310-1 filt -1.jpg
		The flask containing the filtrate was exchanged with a new 50 mL Schlenk flask	filter residue at start of washing.jpg
	15:00 - 15:20	The reaction flask was washed with MeCN (2 * 2.5 mL)	for the first washing: some of the reaction mixture was left in the flask (approx. 3 to 5 mL), since the volume was larger than expected filtrate: AE-310-2 filt -2.jpg
	15:30 - 15:55	The solvents of AE-310-1 was removed under reduced pressure.	Small amount of material was lost due to bubbling up of the mixture when starting to remove the water
	- 16:30	The obtained solid of AE-310-1 was dried under reduced pressure	crude -1.jpg
	16:00 - 16:15	The solvents of AE-310-2 was removed under reduced pressure.	
	- 16:30	The obtained solid of AE-310-2 was dried under reduced pressure	crude -2.jpg
		Both were stored in the dark at - 20 °C	

03.07.	11:00	From now on: AE-310-1 : To the solid THF (15 mL) was added und the obtained mixture was stirred and shaken strongly for approx. 5 min	-1 in THF.jpg
	11:15	n-heptane (15 mL) was added to the mixture	-1 with n-hept stirring.jpg -1 with n-hept after settling.jpg
	11:20 - 11:40	The mixture was filtered through a pore 4 frit according to Protocol - Filtration with frit technique into a 100 mL-Schlenk	
	11:40 - 11:50	The solid was washed with n-heptane (3 * 1 mL)	-1 filter residue wet.jpg 1-filtrate.jpg
	11:50 - 13:40	The solid was dried under reduced pressure	1-dried-product.jpg
08.07	11:55	The product was transfered into a vial in a Schlenk flask	
	14:00	An NMR sample was prepared from AE-310-1 in DMSO-d6 according to Protocol - Preparation of NMR Sample	
		The remaining solid was stored in the dark under argon at -20 °C	
10.07	11:25	From now on: AE-310-2 : To the solid THF (10 mL) was added und the obtained mixture was stirred for approx. 2 min	
	11:27	n-heptane (20 mL) was added to the mixture	
	11:30 - 13:25	The mixture was filtered through a pore 4 frit according to Protocol - Filtration with frit technique into a 50 mL-Schlenk	Very slow filtration
	- 13:32	The solid was washed with n-heptane (3 * 1.5 mL)	much faster than before. It helped, that the filter residue was "dried" before the first washing, meaning, that all liquid was pulled through
	13:35 - 14:45	The solid was dried under reduced pressure	

	15:00	An NMR sample was prepared from AE-310-1 in DMSO-d6 according to Protocol - Preparation of NMR Sample	
		The remaining solid was stored in the dark under argon at -20 °C	
11.07	10:00	An NMR sample was prepared from AE-310-2 in DMSO-d6 according to Protocol - Preparation of NMR Sample	
		The remaining solid was stored in the dark under argon at -20 °C	
19.07	10:00	AE-310-2 was transferred into a screw cap vial in a schlenk Flask under air	
		The flask was set under argon again	
		The remaining solid was stored in the dark under argon at -20 °C	

Analysis

Date	Time	Sample name	Analysis method	Used device	Solvent	Raw Data	Processed Data	Comparative Data	Interpretation
08.07	20:10	AE-310-1	NMR 1H, 31P{1H}quant, 1Hzgespg	IAAC 400 MHz I	DMSO-d6	AE-310-1.zip	AE-310-1_10.nmrium	AE-238-1	Signals as expected, large H2O peak, very few impurities
11.07	12:11	AE-310-2	NMR 1H, 31P{1H}quant, 1Hzgespg	IAAC 400 MHz I	DMSO-d6	AE-310-2-10.zip	AE-310-2_10.nmrium	AE-310-1	Signals as expected, large H2O peak, very few impurities

Product characterization and results

Sample	Mass [mg]	Purity	Mass _{pure} [mg]	Amount [μmol]	Yield [%]	Description	Storage location
AE-310-1	51.81	>97 %	51.81	92.74	18.5	final prod.jpg	freezer drawer A
AE-310-2	25.17	>96 %	25.17	45.06	9	slightly greenish yellow solid	freezer drawer A

With 27.5 % yield lower than without use of MeCN --> maybe not needed?

Future recommendations

Old procedure	Problem	Suggested new procedure
Split mixture in half during workup	/	Do in 1 go, since both fractions were pure

Linked experiments

- AEI-010: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNN})]$ using Ag_2O
- JSC-KS-04: Degassing of n-Heptane
- AEI-024: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{NNN})]$ with 2,6-bis(N,N-diethylaminomethyl) using Ag_2O
- AEI-026: Capillary production
- AEI-048: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{NNN})]$ with 2,6-bis(N,N-diisopropylaminomethyl) using Ag_2O
- AEI-064: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag_2O
- AEI-073: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag_2O
- AE-101: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{PNP})]$ with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$
- AE-102: Degassing of THF
- AE-105: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag_2O
- AE-108: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{PNP})]$ with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$
- AE-114: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag_2O
- AE-121: Degassing of n-Heptane
- AE-145: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag_2O
- AE-157: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{PNP})]$ with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$
- AE-167: Irradiation of PhPDA (365 nm + white LED, gas phase measurement), AE-151, 1.5 mg/mg SDS, 2 mg/mL PhPDA
- AE-198: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ with Ag_2O with different reaction times (1, 2, 5, 24 h)
- AE-KRA-204: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h
- AE-208: Degassing of Milli-Q water
- KRA-080: Degassing of THF
- KRA-104: Degassing of Milli-Q water

- AE-230: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{PNP})]$ with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$

- AE-232: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{PNP})]$ with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$

- AE-238: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h

- AE-246: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-238, 97 % pure) 365 nm and white

- AE-252: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h

- AE-261: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{PNP})]$ with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$

- AE-281: Stability test of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-252) under air and afterwards irradiation with 365 nm and white

Linked resources

Protocol - [Preparation of NMR Sample](#)

Protocol - [Filtration with frit technique](#)

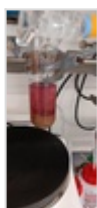
Protocol - [Schlenk Technique](#)

Protocol - [Preparation of capillary for H₂O-NMR](#)

Attached files

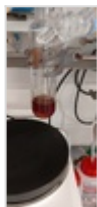
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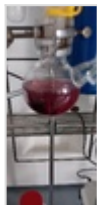
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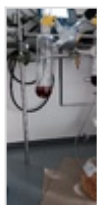
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